RoBenDevs Product Manager Internship
Day 1 (Thursday: 6/11/2026)

Kanban:

What is Kanban?

Kanban is a visual workflow management method that allows teams to optimize and continuously improve the delivery of work by visualizing tasks, limiting Work in Progress (WIP), and focusing on workflow efficiency. It uses visual representations of work to help teams understand what needs to be done, what is currently being worked on, and what has been completed. Unlike Scrum, Kanban does not require fixed-length sprints. Instead, it follows a continuous flow approach where tasks are selected based on the team's available capacity and delivered as soon as they are completed. Kanban is commonly used in software development, IT operations, marketing, customer support, and other areas where teams handle ongoing work and frequently changing priorities.

How Kanban Works:

Kanban works by visualizing the entire workflow, breaking work into smaller tasks, and controlling how much work the team handles at any given time.

The general process involves:

1. **Visualize the Workflow:** Define the different stages through which work passes, such as "To Do," "In Progress," "Review," "Testing," and "Done." These stages are represented as columns on a Kanban board.
2. **Represent Work Using Cards:** Each individual task or work item is represented by a Kanban card. The card contains relevant information such as task description, priority, ownership, and current status.
3. **Limit Work in Progress (WIP):** Establish a maximum limit on how many tasks can be worked on simultaneously within a particular workflow stage. This prevents the team from taking on more work than it can handle.
4. **Pull Work Based on Capacity:** When a team member or workflow stage has available capacity, the team selects the next prioritized task from the previous stage.
5. **Monitor and Improve Workflow:** Use the Kanban board and visual metrics to identify bottlenecks, delays, and inefficiencies. The team continuously adjusts its workflow to improve delivery.

Elements of Kanban:

Kanban is based on a few fundamental principles and practices that help teams maintain an efficient workflow.

1. **Visualize Work:** Every task should be visible to the team through a Kanban board. This creates transparency regarding the current state of work, priorities, and responsibilities.
2. **Limit Work in Progress (WIP):** Teams establish limits on the amount of work that can be active simultaneously. This encourages completing existing work before starting additional tasks.
3. **Manage Flow:** Focus on ensuring that work moves smoothly through the workflow without unnecessary delays or interruptions.
4. **Make Process Policies Explicit:** Define clear rules for how work enters and moves through each workflow stage. For example, a task cannot move to "Done" unless it has passed testing and meets the acceptance criteria.
5. **Implement Feedback Loops:** Regularly review the workflow, discuss blockers, and gather feedback to identify areas that need improvement.
6. **Improve Collaboratively and Experimentally:** Kanban encourages teams to make small, continuous improvements rather than introducing major changes without understanding their impact

Benefits of Kanban:

Kanban provides several advantages by making work visible, controlling workload, and improving the overall delivery process.

Benefit	What it helps teams do	Why it matters
Improved Visibility	Visualize tasks and their current workflow stages	Everyone understands the progress and status of work
Planning Flexibility	Reprioritize upcoming tasks without disrupting ongoing work	Helps teams adapt to changing business requirements
Reduced Bottlenecks	Identify stages where work starts accumulating	Prevents unnecessary delays and improves delivery

Better Workload Management	Limit simultaneous tasks using WIP limits	Reduces multitasking and context switching
Faster Delivery	Improve workflow efficiency and reduce cycle time	Allows teams to deliver value more frequently
Continuous Improvement	Monitor performance using visual metrics	Helps teams identify problems and make data-driven decisions
Improved Collaboration	Establish shared visibility and ownership of work	Encourages team members to work together to complete tasks

Why Should Kanban Be Used?

Kanban is particularly useful when teams need flexibility and continuous delivery rather than working within fixed development cycles.

1. **Better Workflow Visibility:** A Kanban board provides a centralized view of all ongoing work. Product managers, developers, QA engineers, and stakeholders can understand the current status of tasks without repeatedly asking for updates.
2. **Reduced Context Switching:** When team members handle too many tasks simultaneously, they frequently switch between different responsibilities. Kanban's WIP limits encourage the team to focus on completing existing work before starting new tasks.
3. **Early Identification of Bottlenecks:** If tasks start accumulating in the Testing or Review stage, the team can immediately identify that particular stage as a bottleneck and take action to resolve it.
4. **Flexible Prioritization:** Unlike Scrum, Kanban does not require the team to commit to a fixed set of tasks for a sprint. Upcoming work can be reprioritized whenever business requirements change, without necessarily disrupting work already in progress.
5. **Continuous Delivery:** Tasks can be delivered as soon as they are completed and meet the required quality standards. Teams do not need to wait until the end of a sprint to release completed work.
6. **Data-Driven Improvements:** Kanban metrics provide measurable information about how efficiently work moves through the workflow. Teams can use this information to improve their processes instead of relying entirely on assumptions.

Kanban Board:

A Kanban board is a visual workflow management tool designed to help teams visualize work, limit Work in Progress (WIP), and maximize workflow efficiency. A Kanban board represents the different stages of a team's workflow through columns, while individual tasks are represented through cards. As work progresses, cards move from one column to another, providing a clear overview of the team's current workload and progress.

Components of a Kanban Board:

A Kanban board can be broken down into five fundamental components:

1. **Visual Signals (Cards):** Kanban cards represent individual tasks or work items. They provide important information about the task, its owner, priority, and current status.
2. **Columns:** Each column represents a specific stage of the workflow. Common examples include:
    - To Do: Tasks that are ready to be worked on.
    - In Progress: Tasks currently being worked on.
    - Code Review: Tasks waiting for or undergoing code review.
    - Testing: Tasks undergoing QA and validation.
    - Done: Tasks that have been completed and meet the required conditions.
3. **Work in Progress (WIP) Limits:** WIP limits define the maximum number of tasks allowed within a particular workflow stage. They prevent excessive workload and help identify bottlenecks.
4. **Commitment Point:** The commitment point is where the team selects a work item from the backlog and commits to processing it. This marks the beginning of the work's journey through the Kanban system.
5. **Delivery Point:** The delivery point represents the end of the workflow, where the completed product or service is delivered to the customer. The time between the commitment point and delivery point is known as Lead Time.

Example of a Kanban Board:

Backlog	**To Do**	**In Progress (WIP: 2)**	**Review (WIP: 2)**	**Done**
New feature	User Login	Dashboard UI	API Integration	Registration
Payment Gateway	Password Reset	Profile Page		Logout

Search Filter Notifications

The team can pull a new task into the In Progress column only when there is available capacity within its WIP limit.

Kanban Cards:

A Kanban card is a visual representation of an individual work item that needs to be completed. Each card represents a specific task, user story, bug, feature, or any other unit of work that moves through the Kanban workflow. The primary purpose of Kanban cards is to provide all relevant information about a task in one place, allowing team members to understand its requirements, ownership, priority, and current progress.

Important Information in a Kanban Card:

1. **Task Title:** A short description of the work item.
2. **Task Description:** Details regarding what needs to be done and the expected outcome.
3. **Assignee/Owner:** The team member responsible for handling the task.
4. **Priority:** Indicates the importance or urgency of the task.
5. **Status:** Represents the current stage of the task in the workflow.
6. **Due Date:** The expected completion date, if applicable.
7. **Acceptance Criteria:** Defines the conditions that must be satisfied for the task to be accepted.
8. **Dependencies and Blockers:** Identifies external requirements or issues preventing the task from progressing.
9. **Supporting Documents:** Links to relevant designs, requirements, technical documentation, or other resources.

Purpose and Benefits of Kanban Cards:

1. **Improved Visibility:** Team members can quickly understand what needs to be done and the current status of each task.
2. **Clear Ownership:** Every task can have a designated owner, making responsibilities easier to understand.
3. **Smooth Handoffs:** When a task moves between different workflow stages, its requirements and relevant information remain accessible.
4. **Better Tracking:** Cards help teams monitor task progress, identify delays, and understand where work is getting stuck.
5. **Improved Collaboration:** Team members can easily identify tasks that need assistance or are blocked by dependencies.

Work in Progress (WIP) Limits:

Work in Progress (WIP) limits define the maximum number of work items that can exist within a particular stage of a Kanban workflow at a given time. WIP limits are one of the most important aspects of Kanban because they encourage teams to focus on completing existing work rather than continuously starting new tasks. At a fundamental level, WIP limits encourage a culture of "Done" rather than simply starting more work. For example, consider a team that has a WIP limit of 3 for its Development column. This means that the team cannot have more than three active tasks in that column simultaneously. Once one of the existing tasks moves to the next stage, the team can pull another task into Development.

Why Are WIP Limits Important?

1. **Reduce Multitasking:** Working on multiple tasks simultaneously increases context switching and reduces individual productivity. WIP limits encourage team members to focus on fewer tasks.
2. **Identify Bottlenecks:** If a workflow stage reaches its WIP limit, the team can immediately identify that work is accumulating at that particular stage.
3. **Improve Collaboration:** Instead of continuously starting new tasks, team members can help resolve blockers and complete existing work.
4. **Improve Delivery Speed:** Reducing unnecessary multitasking and delays allows work to move through the workflow more efficiently.
5. **Improve Predictability:** By controlling the amount of active work, teams can better understand how much work they can realistically process.

What Happens When a WIP Limit Is Reached?

When a particular workflow stage reaches its WIP limit, team members should avoid pulling additional tasks into that stage.

Instead, they should:

1. Identify why existing tasks are not progressing.
2. Investigate whether there are any blockers or dependencies.
3. Collaborate with other team members to resolve the issues.
4. Help move existing tasks to the next workflow stage.
5. Pull new tasks only when capacity becomes available.

The purpose is to improve the flow of existing work instead of increasing the amount of unfinished work.

Kanban Metrics:

Kanban metrics are measurable data points that help teams understand how efficiently work moves through their workflow.

They provide insights into delivery speed, workload, bottlenecks, and overall workflow performance.

Unlike Scrum, which commonly uses velocity and sprint burndown charts, Kanban primarily focuses on flow-based metrics such as Lead Time, Cycle Time, Throughput, and Work in Progress.

Commonly Used Kanban Metrics:

1. **Lead Time:**
   Lead Time measures the total time taken from the moment a work item enters the committed workflow until it is delivered to the customer. It helps teams understand how long customers have to wait before receiving a completed feature, service, or request. For example, if a customer requests a feature on Monday and the team delivers it on Friday, the Lead Time is 5 calendar days if both endpoints are counted. A shorter Lead Time generally indicates that the team can deliver value to customers more quickly.
2. **Cycle Time:**
   Cycle Time measures the amount of time taken to complete a work item from the moment active work begins until it is finished. For example, if a developer starts working on a task on Tuesday and the task is completed on Thursday, its Cycle Time is approximately 2 days. Cycle Time helps teams understand how efficiently they complete individual tasks and identify work items that are taking longer than expected. A shorter and more consistent Cycle Time generally indicates a smoother workflow.
3. **Throughput:**
   Throughput measures the number of work items completed within a specific period.
   For example, if a development team completes 15 tasks within one week, its weekly Throughput is 15 work items. It helps teams understand their delivery capacity and estimate how many tasks they can realistically complete within a given period. Unlike Velocity, Throughput counts completed work items rather than measuring their estimated effort in story points.
4. **Work in Progress (WIP):**
   WIP measures the total number of work items that are currently being processed within the workflow. For example, if a team has 3 tasks in Development, 2 in Code Review, and 2 in Testing, its total WIP is 7 tasks. Tracking WIP helps teams understand their current workload and determine whether too much work is being handled simultaneously.
5. **Flow Efficiency:**
   Flow Efficiency measures the proportion of time spent actively working on a task compared to the total time the task spends within the workflow. It helps teams understand how much time is spent performing actual work versus waiting for approvals, reviews, dependencies, or other activities.

   Formula:

Flow Efficiency = (Active Working Time / Total Lead Time) × 100

For example, if a task takes 10 days from commitment to delivery but only 4 days are spent actively working on it, its Flow Efficiency is 40%.
A low Flow Efficiency may indicate unnecessary waiting periods, handoff delays, or workflow bottlenecks.

Visual Metrics in Kanban:

Kanban uses visual reports and charts to make workflow performance easier to understand. Two commonly used visual reports are Control Charts and Cumulative Flow Diagrams.

1. **Control Chart:**
   A Control Chart displays the Cycle Time of individual work items over a period of time.
   It helps teams understand how long tasks usually take to complete and whether their delivery time is consistent.
   For example, if most tasks take between 3 and 5 days but a few take 12 days, the team can investigate what caused those unusually long delays.
   Benefits:
   - Identifies tasks that take longer than expected.
   - Helps understand the consistency of Cycle Time.
   - Highlights unusual delays and potential workflow issues.
   - Helps teams estimate future delivery timelines more realistically.
2. **Cumulative Flow Diagram (CFD):**
   A Cumulative Flow Diagram is a visual chart that displays the number of work items within each workflow stage over time.
   Each colored band represents a particular workflow stage, such as To Do, Development, Review, or Done.
   It helps teams understand how work accumulates and moves through the workflow.
   For example, if the Development band continuously becomes wider while the Review band remains relatively narrow, it may indicate that tasks are entering Development faster than they are leaving it.
   This suggests a potential bottleneck that needs attention.
   Benefits:
   - Identifies workflow bottlenecks.
   - Shows whether work is accumulating in a particular stage.
   - Helps monitor WIP over time.
   - Provides visibility into the overall stability of workflow.
   - Helps teams understand whether work is flowing consistently.

How Visual Metrics Simplify Kanban:

Visual metrics turn workflow data into understandable information, allowing teams to identify problems without manually inspecting every task.

For example:

1. If the Cycle Time increases, the team can investigate why tasks are taking longer to complete.
2. If the CFD shows work accumulating in Testing, the team can identify QA as a potential bottleneck.
3. If Throughput decreases, the team can investigate whether there are resource constraints, blockers, or changes in task complexity.
4. If WIP remains consistently high, the team can review whether it is taking on more work than its actual capacity.
5. If Flow Efficiency is low, the team can investigate unnecessary waiting periods and reduce handoff delays.

The overall purpose of Kanban metrics is not simply to measure team performance but to identify inefficiencies, improve predictability, and continuously optimize the workflow.

Kanban Best Practices:

1. **Make Blockers Visible Early:** Identify and communicate blocked tasks as soon as possible so that the team can resolve them before they cause further delays.
2. **Keep WIP Limits Realistic:** Establish WIP limits based on the team's actual capacity and adjust them as the workflow changes.
3. **Review Bottlenecks as a Team:** Regularly inspect the Kanban board and identify stages where work frequently accumulates.
4. **Encourage Collaboration:** Team members should help move existing work forward instead of focusing exclusively on their individually assigned tasks.
5. **Define Clear Workflow Policies:** Establish clear conditions for when a task can enter or leave a particular workflow stage.
6. **Use Metrics for Continuous Improvement:** Regularly review Cycle Time, Lead Time, Throughput, and other metrics to identify patterns and make informed improvements.

Scrumban:

Scrumban is a hybrid Agile approach that combines elements of Scrum and Kanban.It combines the structured planning and coordination practices of Scrum with the flexibility and continuous workflow management of Kanban.Scrumban is particularly useful for teams that want the predictability of Scrum but also need the flexibility to manage changing priorities and ongoing work.Unlike traditional Scrum, Scrumban does not necessarily require fixed-length sprints. Teams can use continuous workflow management while retaining selected Scrum practices such as regular planning meetings, reviews, and retrospectives. Scrumban uses Kanban boards and WIP limits to visualize work and control how much work the team handles simultaneously.

How Scrumban Works:

1. **Maintain a Product Backlog:** Similar to Scrum, the team maintains a prioritized backlog containing features, enhancements, bugs, and other requirements.
2. **Visualize Work Using a Kanban Board:** Tasks are represented through cards and moved across different workflow stages as work progresses.
3. **Use WIP Limits:** The team establishes limits for active work to prevent excessive multitasking and identify bottlenecks.
4. **Plan Work Based on Capacity:** Instead of necessarily committing to a fixed set of tasks for an entire sprint, the team selects new tasks as capacity becomes available.
5. **Conduct Regular Reviews and Retrospectives:** The team reviews completed work, gathers stakeholder feedback, and identifies opportunities for continuous improvement.

Important Characteristics of Scrumban:

1. **Hybrid Approach:** Combines Scrum's planning structure with Kanban's flow-based workflow management.
2. **Flexible Planning:** Teams can adjust priorities and select work based on available capacity.
3. **Visual Workflow:** Uses Kanban boards to maintain transparency regarding task progress.
4. **WIP Limits:** Controls the amount of work being handled simultaneously.
5. **Continuous Improvement:** Encourages regular feedback and process optimization.
6. **Flexible Delivery:** Work can be completed and delivered as soon as it is ready, depending on the team's release process.

When Should Scrumban Be Used?

Scrumban is particularly useful when:

1. A team wants to transition from Scrum to Kanban gradually.
2. The team needs regular planning and review activities but does not want strict sprint commitments.
3. Priorities change frequently, making fixed sprint planning difficult.
4. The team handles both planned development work and unexpected requests or bug fixes.
5. The team wants to improve workflow efficiency without completely removing its existing Scrum practices.

Scrum vs. Kanban vs. Scrumban:

Although all three approaches are used in Agile environments, they differ in how they organize, plan, and deliver work.

1. **Scrum:** Scrum is an iterative Agile framework that organizes work into fixed-length sprints. Teams select work during Sprint Planning, work towards a Sprint Goal, and review their progress at the end of each sprint.
2. **Kanban:** Kanban is a flow-based workflow management method. Teams visualize work, establish WIP limits, and continuously pull new tasks based on available capacity.
3. **Scrumban:** Scrumban combines the structured planning practices of Scrum with the continuous workflow management of Kanban.

Comparison Between Scrum, Kanban, and Scrumban:

Criteria	**Scrum**	**Kanban**	**Scrumban**
Approach	Iterative framework	Flow-based method	Hybrid approach
Workflow Style	Fixed-length sprints	Continuous workflow	Flexible, flow-based workflow
Planning Style	Sprint Planning	Continuous prioritization and replenishment	Regular planning based on capacity
Delivery Style	Usually organized around sprint increments	Continuous delivery	Flexible delivery
Work Management	Product Backlog and Sprint Backlog	Kanban board and prioritized work queue	Backlog and Kanban board
WIP Limits	No prescribed Kanban-style WIP limits	Essential practice	Commonly used

Roles	Product Owner, Scrum Master, Developers	No prescribed roles	Flexible; often retains existing Scrum roles
Meetings	Defined Scrum events	Flexible feedback and review meetings	Selected Scrum events and regular workflow reviews
Change Management	Changes should not endanger the Sprint Goal	Priorities can change as work moves through the system	More flexible changes based on team policies
Main Metrics	Velocity, Sprint Burndown, Sprint Goal Success	Lead Time, Cycle Time, Throughput, WIP	Cycle Time, Throughput, WIP, and selected sprint metrics
Best For	Teams working towards defined goals in structured cycles	Teams handling continuous incoming work	Teams needing both planning structure and workflow flexibility

When to Use Each Method:

1. **Scrum:**
    - When the team benefits from structured planning and fixed-length development cycles.
    - When the team works towards clearly defined Sprint Goals.
    - When requirements can be organized into manageable increments.
    - When the team benefits from defined accountabilities and regular Scrum events.
    - When stakeholders need predictable review and feedback intervals.
2. **Kanban:**
    - When the team handles a continuous flow of incoming requests.
    - When priorities change frequently and flexibility is important.
    - When the team wants to improve workflow efficiency and reduce bottlenecks.
    - When work does not naturally fit into fixed-length sprints.
    - When continuous delivery is preferred over sprint-based delivery.
3. **Scrumban:**
    - When a team wants Scrum's planning structure but needs more flexibility.

- When the team frequently receives unexpected tasks alongside planned development work.
- When transitioning from Scrum to Kanban without completely changing the existing process.
- When the team wants to maintain regular planning and review meetings while controlling work through WIP limits.
- When strict sprint commitments are not suitable for the team's workflow.

Practical Example:

Consider three software development teams working on different types of products:

1. **Team A — Scrum:** A team developing a new ERP system with clearly defined features and development goals. They can organize requirements into two-week sprints, establish Sprint Goals, and demonstrate completed features at the end of each sprint. Scrum would be suitable here.
2. **Team B — Kanban:** An IT support team handling continuous incoming requests, bug reports, and technical issues. Since requests arrive unpredictably and priorities change frequently, Kanban allows the team to prioritize work dynamically and deliver tasks as soon as they are completed.
3. **Team C — Scrumban:** A team maintaining an existing SaaS product while also developing new features. The team needs regular planning meetings for feature development but must remain flexible enough to handle unexpected production bugs and urgent requests. Scrumban would be suitable here.

Final Distinction:

- Scrum focuses on completing planned work within fixed-length iterations.
- Kanban focuses on improving the continuous flow of work and reducing delivery time.
- Scrumban combines structured planning with flexible workflow management.

The selection depends on the nature of the work, how frequently priorities change, and how much structure the team needs to m